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Widespread Cenozoic (?) remagnetization in Thailand and its implications for the India–Asia collision

Chen Yan and Vincent Courtillot

Laboratoire de Paléomagnétisme et Géodynamique, Institut de Physique du Globe de Paris, 4, Place Jussieu, 75252 Paris Cedex 05 (France)

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Paleomagnetic study of 13 sites from the Khorat plateau in Thailand spanning ages from the Devonian to the Jurassic, and reappraisal of published results from the Triassic to the Cretaceous show that most of the Indochina block suffered complete remagnetization after the Triassic and possibly after the Cretaceous, most likely at the time of the collision of India and Eurasia. The paleodirection associated with this remagnetization is consistent with the contention that Thailand was a part of Eurasia at that time and that it has suffered some 20° of clockwise rotation subsequently. Also consistent is the observed azimuthal difference of Mesozoic fold axes on either side of the Red River fault, as is the early geometry of and sense of motion on this fault. Re-interpretation of late Neogene data suggests that rotation may have continued into the Neogene. The remagnetization event prevents the determination of paleopositions of Thailand prior to the Cenozoic, but carries great interest in the frame of the extrusion tectonics concept.

1. Introduction

It is now clear that much of the present morphology and tectonics of South-East Asia has been shaped by the collision of India against Asia (e.g. [1–3]). Until recently, the age of the collision was regarded as being Eocene (about 50 Ma [4]), but it has recently been suggested that first contact could have been established as early as the Cretaceous–Tertiary boundary (about 65 Ma [5]). The extent to which the collision has controlled the Cenozoic tectonics of Asia as far as northern China and the Pacific margin remains a matter of debate. However, a central prediction of the extrusion tectonics model of Tapponnier et al. [1] is the large-scale rotation of much of Indochina (Sundaland) during the first phases of the collision. Achache et al. [6] reviewed paleomagnetic work available at that time and concluded that indeed a large clockwise rotation of Indochina was supported. In an attempt to better constrain the pre-collision history of this block, Achache and Courtillot [7] reported new paleomagnetic results from the upper Triassic formations of the Khorat Group in Thailand. The observation of both normal and reversed polarities as well as an apparently positive foldtest on five sites, added to earlier reports of magnetostratigraphies from middle Triassic to Cretaceous [8,9], led Achache and Courtillot to conclude that they had indeed been successful in isolating a primary Triassic magnetization. Utilizing this result, a Mesozoic apparent polar wander path for Indochina was further developed, with the conclusion that the block had remained at an approximately constant paleolatitude for the last 200 Ma. The preliminary nature of these results, however, was also pointed out [7]. We have since added to the analysis two Devonian sites, four Permian sites, another Triassic site and a Jurassic site, which increases the total number of paleomagnetic samples from an originally very low value of 26 to a more acceptable one of 102. This combined data set is presented here.

The samples are all from the same 1981 collection along the western and northern edges of the Khorat plateau (Fig. 1). The Khorat Group consists of a 4000 m thick series of mostly continental sediments ranging from upper Triassic to lower Cenozoic. The series is rather gently folded except on the western margin where dips may reach 25°. It lies unconformably over folded Paleozoic formations exposed on the edges of the plateau. The

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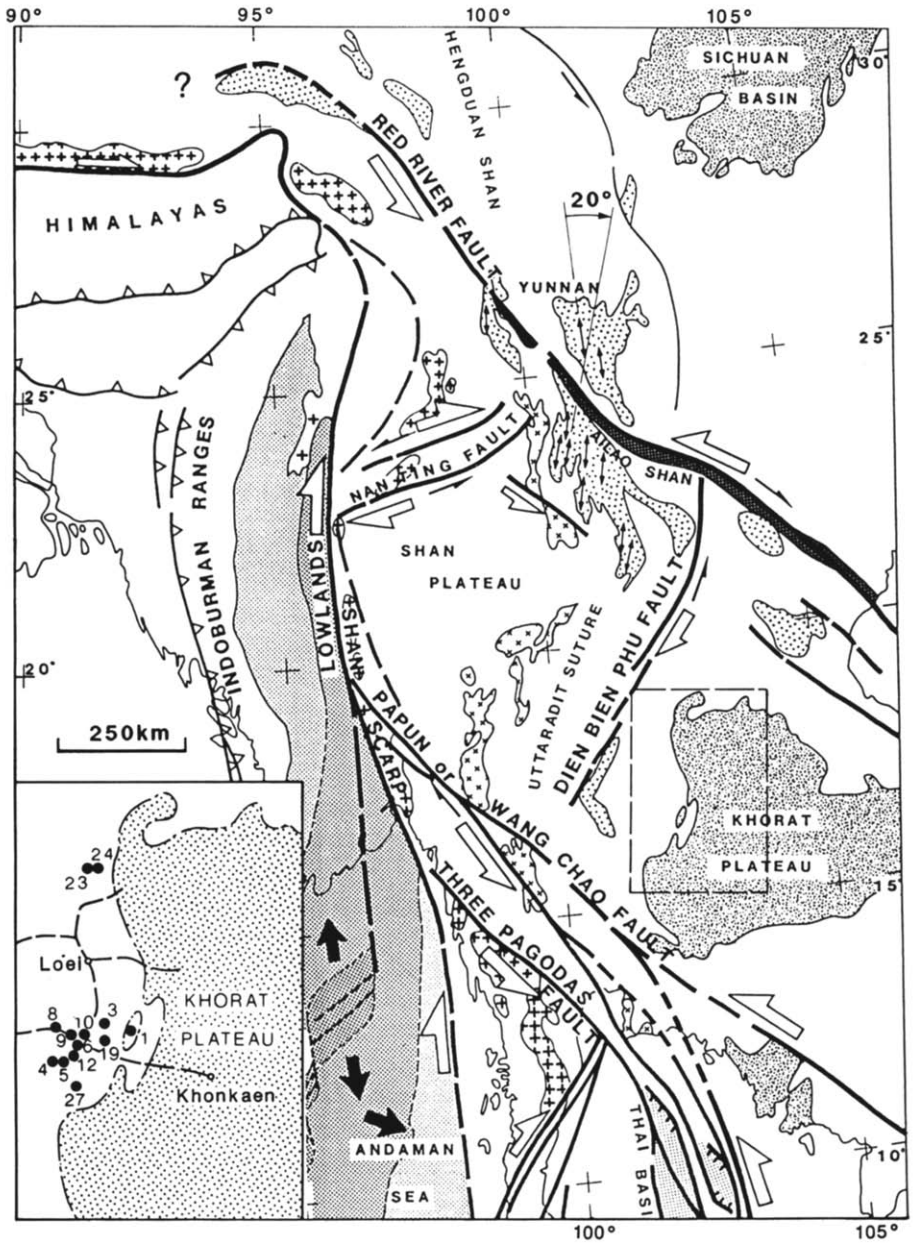


Fig. 1. Simplified structural setting of the Indochina block, with the South China block to the north of the Red River fault and the Indian plate to the west of the Andaman Sea. Paleomagnetic sampling sites on the edges of the Khorat plateau are indicated by dots in the enlarged inset (lower left corner). Note fold axes in Mesozoic continental sediments on either side of the Red River–Ailao Shan fold system (small double arrows). Large open half arrows indicate sense of motion on major faults in the first phase of the collision, whereas thin half arrows indicate recent (present) motion. (From Tapponnier et al. [2].)

Devonian sites consist of andesite (24) and greenish sandstone (23), while the Permian sites (6,9,10,12) are all bluish to gray (sometimes pale pink) limestone rich in *Fusulina* fossils. Further, the Triassic Huai-Hin Lat Formation begins with a basal sandstone and conglomerate (5). Other sites include grey sandstones (4,27) and limestones (3 and 19; 8 is somewhat bituminous) while the single Jurassic site (1) is a pink sandstone. Clearly, with the possible exception of the Triassic, these

sites are not numerous enough to provide meaningful averages for each separate stage. Nonetheless, we now present evidence for pervasive remagnetization of these rocks, and show that the total of 13 sites may provide a good estimate of the direction of this remagnetization.

2. Paleomagnetic results

Measurements were done in the mu-metal/steel shielded room of the Paleomagnetic laboratory at IPG in Paris, with a three-component CTF cryogenic magnetometer. The natural remanent magnetization (NRM) ranged in strength from 10^{-4} to 10^{-1} A m^{-1}. Thirty-one sites had actually been sampled in Thailand in 1981, but only thirteen yield interpretable paleomagnetic results. This is a large rejection rate, related to the high degree of alteration and poor preservation of most samples. For completeness, some examples of demagnetization diagrams for the eighteen sites not reported in detail here are given in fig. 2. They show weak magnetization, at or below the resolution of the magnetometer, or completely erratic demagnetization paths, or overwhelming remagnetization during thermal treatment.

For the thirteen remaining sites, thermal demagnetization was often found to be more efficient than af in revealing discrete magnetic components, and was used essentially throughout. A comparison of behaviour of samples under thermal vs. af treatment was given in Achache and Courtillot [7, fig. 3]. Af demagnetization in general revealed two distinct components, but their coercivity spectra often strongly overlapped (except for site 8). Typical demagnetization diagrams are shown in Fig. 3. As can be seen, a rather large normal overprint is removed by 120 °C to 300 ° or 400 ° C. After removal of this overprint, a second component appears, in general with univectorial

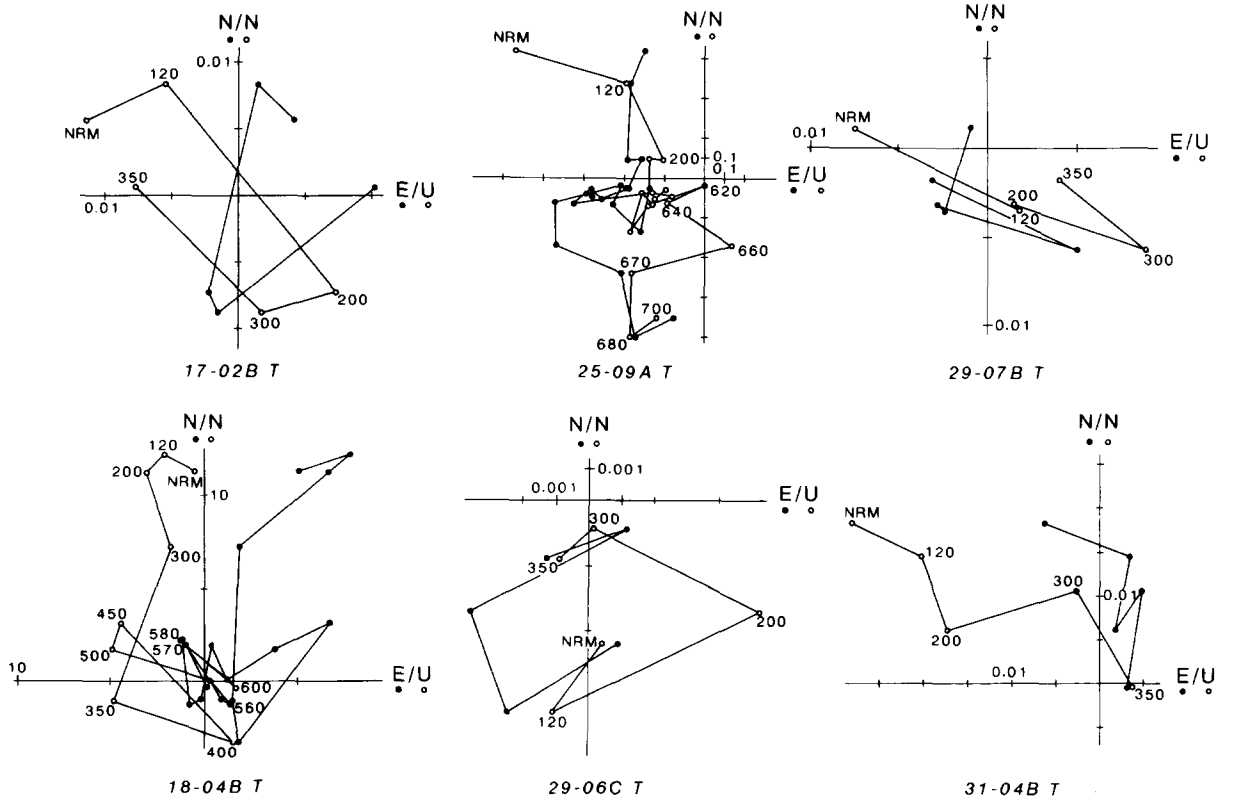


Fig. 2. Orthogonal vector projection (in units of mA/m) of thermal demagnetization for typical samples of the eighteen sites from which no meaningful magnetic component could be extracted. All are in geographic coordinates. Solid symbols are in the vertical NS/Up-Down plane, and open symbols in the horizontal NS/EW plane.

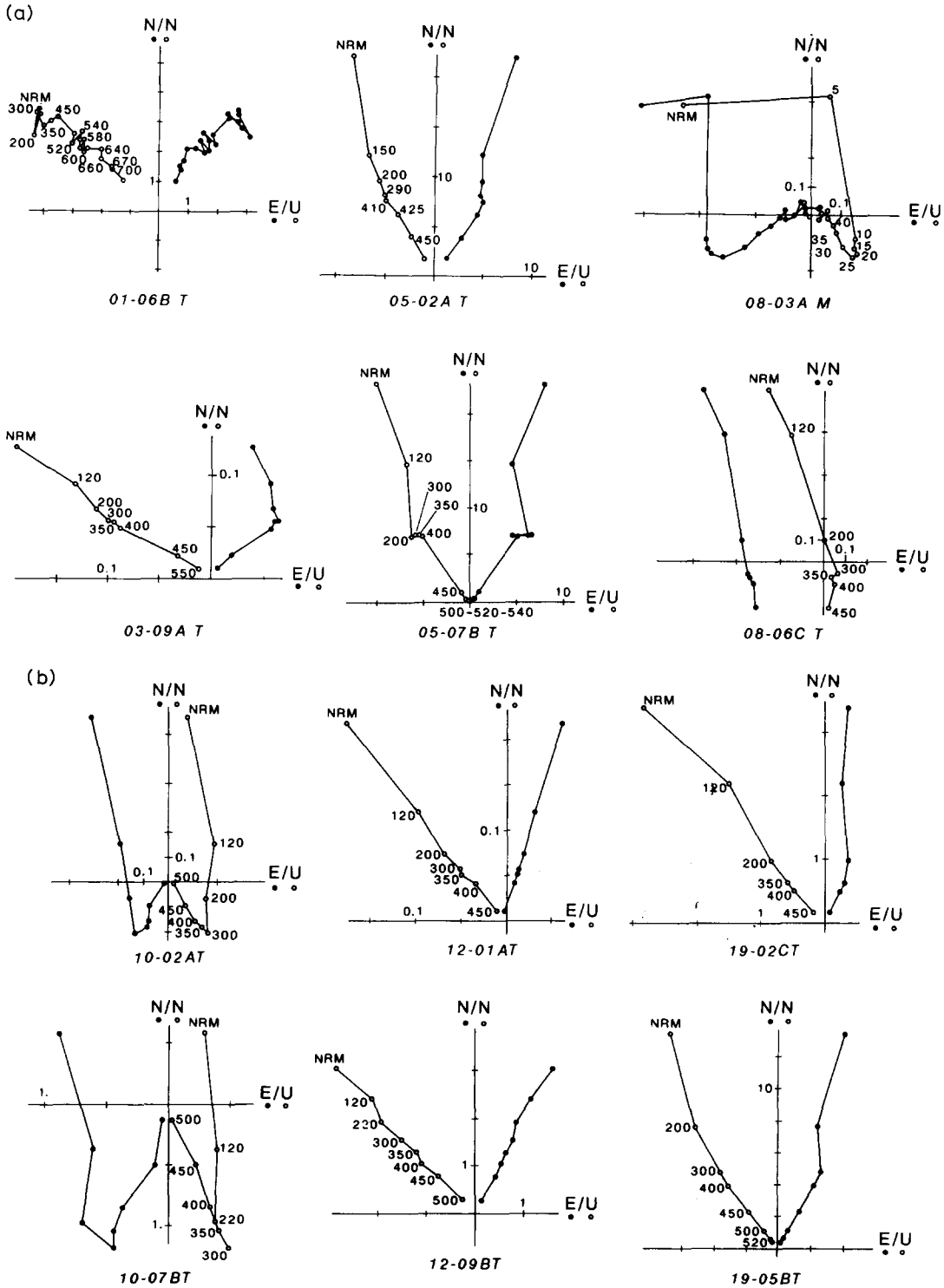


Fig. 3. Orthogonal vector projection (in units of mA/m) of mostly thermal (T) and one af (M) demagnetizations of selected specimens from the Khorat plateau, all in geographic (in situ) coordinates. Solid symbols are in the NS/Up-Down plane, and open symbols in the horizontal NS/EW plane.

decay to the origin by 450–550 °C for 11 out of the 13 sites. The same component appears between 450 ° and 700 °C in site 1. In site 8, the overprint is not removed until 450 °C, when strong laboratory remagnetization (not shown in Fig. 3a) takes place. A clear reversed component is suggested beyond 450 °C, but is only well isolated with af demagnetization between 25 and 50 mT (Fig. 3a, sample 08-03 AM). This indicates that in most sites some form of (titano) magnetite is a likely carrier of the characteristic magnetization; some hematite is present in the pink sandstone of site 1; the very weak reversed component in site 8, a bituminous limestone, might be carried by sulfides, unblocking taking place under 280 °C.

Principal component analysis of the low temperature overprint showed that it is scattered close to the present field direction in geographic coordi-

TABLE 1

Paleomagnetic results for the Khorat Plateau (this study)

Age	Site	n	Geographic coordinates				Stratigraphic coordinates				K_g/K_s
			$D(°)$	$I(°)$	K_g	$\alpha_{95}(°)$	$D(°)$	$I(°)$	K_s	$\alpha_{95}(°)$	
J	1	13	31.8	45.3	166	3.2	25.4	35.8	94	4.3	
Tr	3	13	36.7	46.2	68	5.1	41.6	54.6	70	5.0	
Tr	4	12	32.8	31.5	11	13.6	35.6	41.8	11	13.6	
Tr	5	14	32.5	29.7	78	4.5	36.6	49.1	78	4.5	
Tr	8	5	260.4	−22.6	12	22.7	273.0	−55.6	12	22.7	
Tr	19	9	29.1	35.9	58	6.8	37.0	33.1	59	6.7	
Tr	27	3	30.9	42.6	76	14.2	30.6	55.4	78	14.1	
Mean		6	40.8	36.2	19	15.9	43.6	50.0	23	14.1	0.83
		5	32.3	37.2	120	7.0	36.4	46.9	70	9.3	1.71 * excluding site 8
P	6	9	25.5	45.4	68	6.3	66.1	30.0	69	6.2	
P	9	2	32.2	27.6	–	–	54.0	25.1	–	–	
P	10	8	200.1	−19.8	48	8.1	255.9	−29.3	74	6.5	
P	12	8	24.3	40.5	377	2.9	42.8	38.9	178	4.2	
Mean		4	25.5	33.4	42	14.4	60.1	31.4	36	15.4	1.17
		3	27.6	37.9	68	15.0	54.7	31.7	45	18.5	1.51 excluding site 10
D	23	3	39.5	46.2	120	11.3	62.0	−12.1	120	11.3	
D	24	3	183.6	−39.4	99	12.5	192.6	8.0	16	31.6	
Mean		2	20.6	44.2	–	–	36.0	−11.2	–	–	
Mean of sample directions											
		102	29.7	38.2	22	3.1	44.4	40.2	11	4.3	2.00 **
		89	29.5	39.5	31	2.7	38.7	38.3	15	4.1	2.07 ** excluding sites 8 and 10
Mean of site means											
		13	32.3	38.3	25	8.1	43.9	31.6	6	17.1	4.17 **
		12	31.9	39.9	100	4.6	38.2	28.4	7	18.4	14.28 ** excluding site 8
Mean of age-group means											
		4	28.2	41.2	195	6.6	38.3	26.7	9	32.7	21.67 **

J = Jurassic, Tr = Triassic, P = Permian, D = Devonian. Site number: see Fig. 1. n = number of specimens or sites. K_g, K_s = Fisher's parameter before and after bedding correction, respectively.

* Indicates significant negative fold tests (at the 95% confidence level).

** Indicates significant negative fold tests (at the 99% confidence level).

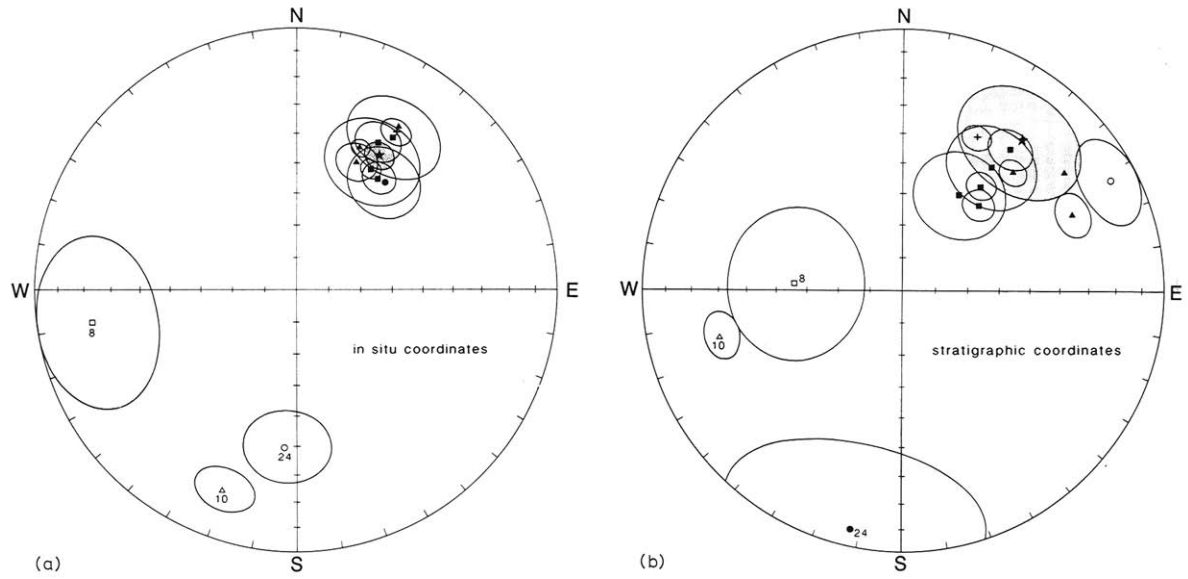


Fig. 4. Equal-area projection of site mean directions, before (a) and after (b) bedding correction, with circles of 95% confidence. Note that there are only two specimens available for site 9, hence no confidence interval. For site locations see Fig. 1, and for ages see Table 1. Symbols are as follows: circle = Devonian, triangle = Permian, square = Triassic, cross = Jurassic, star = overall site-mean ($N = 12$ sites; site 8 excluded). Closed symbols are in the lower hemisphere and open symbols in the upper hemisphere.

nates, hence likely to be a recent component, perhaps of viscous origin. The results of interest here concern the medium-to-high-temperature component, for which statistical results are listed in Table 1. 95% confidence intervals range from 3° to 23° with an average of 9°. Ten out of thirteen sites have a normal polarity. The reversed sites are the Devonian andesite (24), Permian limestone (10) and Triassic bituminous limestone (8). The magnetization in site 8 was very weak and could not always be extracted, hence the large α_{95} of 23°. The normal directions are well grouped in geographic coordinates (Fig. 4a) and the reversed directions 10 and 24 are reasonably antipodal to the normal ones. Ill-determined direction 8 is clearly removed from all others but cannot be given large weight of confidence for the reasons outlined above. When corrected for bedding, the characteristic directions tend to diverge significantly (Fig. 4b). Specifically, the α_{95} associated with the overall mean for all sites rises from 8.1° to 17.4° ($N = 13$) after correction, a negative "foldtest" at the 99% confidence level [10]. Strictly speaking, this is not a legitimate foldtest, since sites extend from the Devonian to the Jurassic, and are not expected to have had uniform original magnetizations. When site mean directions are averaged separately for each age group, the numbers of sites and the differences in strike and dip (Fig. 5) are in general too small to allow a significant foldtest (Table 1). The poles to bedding (Fig. 5) tend to be clustered for each age group. Also,

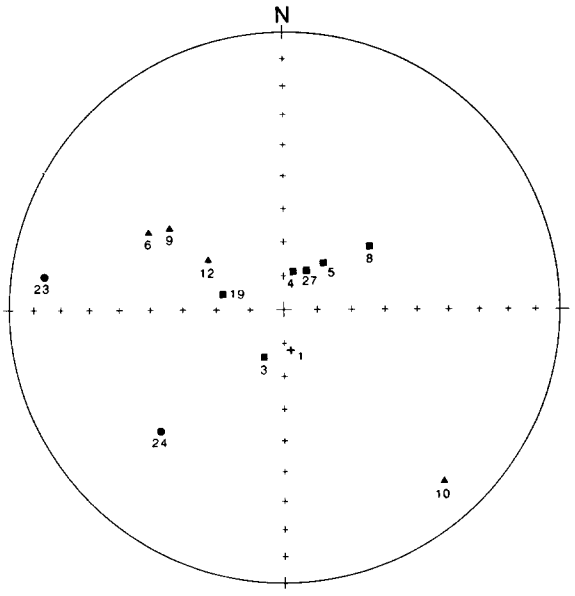


Fig. 5. Poles to bedding planes of paleomagnetic sampling sites. Site numbers are indicated. Age symbols as in Fig. 4.

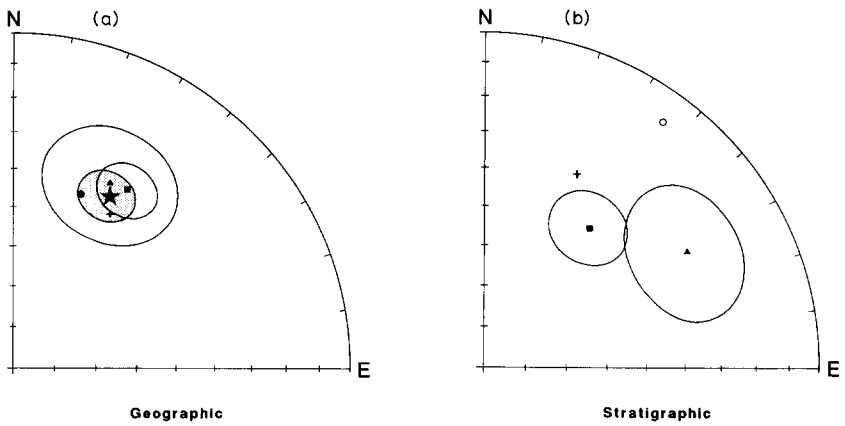


Fig. 6. Equal-area projection of age-group-mean directions, before (a) and after (b) bedding correction, with circles of 95% confidence. Age symbols as in Fig. 4. The overall mean ($N = 4$ age-groups) is indicated by a star, only prior to bedding correction.

structural differences between the age groups are sufficiently large (Fig. 5) to allow a significant, negative "foldtest" at all statistical mean levels (Fig. 6 and Table 1). The clustering of in situ directions (Fig. 4a) strongly suggests that magnetization was acquired after the tectonic events that induced folding.

A further indication that remagnetization indeed occurred is given by results from site 5 (Triassic basal conglomerate), in which samples were drilled both from the coarse-grained Triassic sandstone matrix and pebbles of Permian limestone. The characteristic directions from all cores are in close agreement, and the overall mean has an α_{95} of 4.5°. Clearly, this implies a negative conglomerate test and further makes it doubtful that the underlying Permian rocks now carry a primary magnetization. Based on a marginally positive foldtest, Achache and Courtillot suggested that the magnetization might be Triassic. The test [7, fig. 4], however, relied almost entirely on the poorest determined direction, that of site 8. Given our additional data, the Triassic sites no longer provide a positive foldtest; on the contrary, the α_{95} for Triassic sites increases when site 8 is excluded and the foldtest is negative at the 95% confidence level (Table 1).

There is therefore little doubt that all of the sites reported in this paper and in Achache and Courtillot [7] have been remagnetized after the Triassic. The single Jurassic site reported here suggests that the remagnetization event may be considerably younger (Figs. 4 and 6).

3. Discussion

Significant amounts of paleomagnetic data have been published on rocks from Thailand. Barr et al. [11] reported a total of 40 sites of middle Triassic to upper Jurassic age. Bunopas [8] reported 24 sites of upper Triassic to Cretaceous age, and Maranate [9], and Maranate and Vella [12] 71 sites of middle Triassic to Cretaceous age. All of these results are summarized in Table 2. Although mean directions tend to agree, there is wide scatter. It is particularly unfortunate that none of the above authors have listed directions prior to bedding correction, or given bedding parameters. It is interesting that the many Jurassic results of Bunopas, for instance, are spread in the first quadrant, with very few reversed results, in a pattern similar to that found in our Fig. 4b. This suggests that all of these Jurassic rocks have indeed experienced the same remagnetization event that we identify. It is also significant that the 12 Cretaceous sites of Bunopas and Maranate again show similar behaviour, with a declination ranging from 10 to 50°. These data are associated with rather large confidence intervals, which again are not listed prior to bedding correction. When all these data are considered, we argue that there is a strong indication that all Paleozoic and Mesozoic rocks from the Khorat plateau may have undergone the same widespread and complete remagnetization after the Cretaceous subsequent to folding.

The only significant tectonic event known to have occurred in the vicinity of Thailand since the

TABLE 2

Summary of published paleofield directions from the Khorat Plateau

Age	N (n)	Stratigraphic coordinates				Reference
		D (°)	I (°)	K_s	α_{95}	
Tu	29 (?)	4	28	17	7	[15]
Tu	5 (24)	200	−30	148	5	[22]
K	7 (21)	37	32	63	9	[12]
J	1 (13)	25	36	–	–	this study
Jm	14 (42)	31	41	10	13	[12]
Jm	16 (48)	33	35	23	8	[12]
Jl	16 (44)	39	36	19	9	[8]
Jl	15 (45)	33	29	31	7	[12]
Tr.u-Jl	9 (27)	34	41	20	12	[12]
Tr.u-Jl	8 (24)	30	20	11	18	[8]
Tr.m?	10 (30)	18	21	13	14	[12]
Tr.m-Jl	6 (19)	36	39	32	12	[11]
Tr-J	8 (28)	26	36	15	15	[11]
Tr.u	5 (51)	44	50	23	14	this study
P	4 (27)	55	32	45	18	this study
D	2 (6)	36	−11	–	–	this study

T = Tertiary, K = Cretaceous, J = Jurassic, Tr = Triassic, P = Permian, D = Devonian; u = upper, m = middle, l = lower. N and n are numbers of sites and samples respectively.

Note that all these directions have apparently been corrected for bedding although, except for the present study, in situ directions are unfortunately not available. This table is intended to demonstrate the general resemblance of bedding corrected directions, at least up to the Cretaceous, and the rather large 95% confidence intervals, but of course the directions listed here should not be used further, since they are interpreted as a post-folding remagnetization erroneously bedding-corrected. For this reason pole positions are not computed.

beginning of the Mesozoic is the collision of India and Eurasia, prior to which Indochina formed the stable southern margin of Eurasia (at least since its Triassic(?) suturing with South China; see for instance Buffetaut [13]). Besse [14] has recently proposed a detailed, synthetic apparent polar wander path (APWP) for Eurasia, against which the remagnetization which we observe can be compared (Fig. 7). the paleolatitude of $22.0 \pm 3.3°$ associated with remagnetization is consistent with the expected Eurasian paleolatitude since 70 Ma, except perhaps for a short interval of time around 30 Ma. The declination on the other hand is quite discrepant. Depending on the time at which the remagnetization was acquired, this discrepancy amounts to a clockwise rotation increasing from $20 \pm 5.5°$ at 60 Ma, to $32 \pm 4.5°$ at present (Fig. 7). Hence, the more recent the event, the larger the rotation. Tapponnier et al. [1,2] and Peltzer and Tapponnier [3] have put forward arguments in favour of an early extrusion of Indochina, taking place between the onset of generalized collision at about 50 Ma and the end of opening of the South China Sea at 17 Ma. Major motion between the Indochina and South China blocks is argued to have been taken up along the (then) left lateral Red River fault. Tapponnier et al. [2] further point out that the Mesozoic sedimentary cover of Indochina and South China near 25° N, 100° E has been folded, most likely during the early phases of the collision, probably in part prior to major

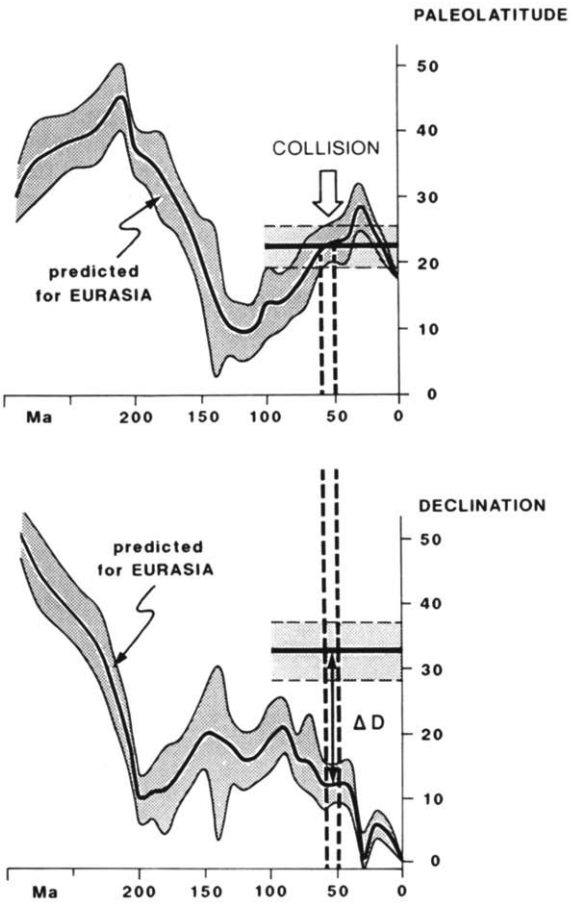


Fig. 7. The paleolatitude and declination of the Khorat remagnetization is compared to directions predicted from the Eurasian APWP of Besse [14] for a reference point at 17° N, 103° E. Paleolatitudes agree from 70 Ma to the present (except around 30 Ma) but declinations are significantly different. The declination difference ΔD is outlined in the hypothesis of a 60–50 Ma remagnetization.

offset on the Red River fault (Fig. 1). The general fold direction is consistent with left lateral motion on the fault. Moreover fold axes on the south side of the fault (Indochina) are rotated clockwise by 20° with respect to fold axes north of the fault (South China) (Fig. 1).

This rotation is consistent with the rotation of the remagnetization, if it took place at the onset of the collision (more or less coeval with the folding) as well as with the overall sense of rotation of Indochina in the early phase of indentation (when the left lateral Red River fault was curved towards the South). This interpretation would imply that widespread remagnetization occurred immediately after the tectonic phases which have folded Thailand formations as a distant consequence of the collision.

It would be desirable to have direct paleomagnetic data from Thailand associated with ages intermediate between the onset of the collision and the present. Such data are not available for the Paleogene. However, what appear to be very young Neogene data have recently been published by McCabe et al. [15]. These data are from basaltic flows covering much of Thailand and dated between 0.6 and 11.3 Ma. McCabe et al. find that the mean direction for 28 flows from the Khorat plateau (eastern Thailand) is indistinguishable from the present dipole field direction, whereas that for 18 flows from central and western Thailand display a declination deflected significantly clockwise by some $24 \pm 8°$. McCabe et al. [15] interpret their results in terms of a very recent opening of the basins of central Thailand about a nearby pole in northern Thailand. It so happens that the few available ages from the eastern sector range from 0.4 to 3.3 Ma, whereas those from the western sector are between 3.1 and 11.3 Ma. The reliability of these few K-Ar ages may be in doubt: the basalts occur in a tropical climate where alteration is pervasive, and there are indications that K-Ar ages may be extensively contaminated by argon loss, as is for instance the case in the nearby Deccan Basalts [16]. There is therefore a possibility that the eastern sites of McCabe et al. [15] are significantly younger than those in the center and west of the country. The older rocks might be related to early opening of the north-south trending basins of central Thailand, and volcanism might have subsequently shifted towards the east. Although this hypothesis remains to be tested, by careful ^{39}Ar-^{40}Ar dating of the basalts, we see that a 20 Ma age of the western volcanism would be consistent with a $\sim 10°$ clockwise rotation of the whole of Thailand. That volcanism shifting in both time and space may have recorded the latest stages of extrusion of Indochina and South China is an interesting possibility.

4. Conclusion

We conclude from our study of 13 sites coming from the Khorat plateau in Thailand and spanning ages from the Devonian to the Jurassic, and from a reappraisal of published results from the Triassic to the Cretaceous that much of Indochina seems to have suffered complete remagnetization after the Triassic and possibly after the Cretaceous. This remagnetization, which could have taken place shortly after gentle folding of the Thai formations at the time when the collision of India against Eurasia became generalized (50 Ma), is reminiscent of low-temperature, continent-wide chemical remagnetization events that occurred after the late Paleozoic orogenies in North America [17] or Europe [18]. Zhao and Coe [19] and Kent et al. [20] have found that many Paleozoic formations in South and Eastern China suffered pervasive remagnetization in late Mesozoic to recent times. Zhao and Coe invoke a thermal event related to the Yenshanian orogeny, whereas Kent et al. suggest a chemical or diagenetic origin. The latter mechanism, perhaps related to widespread fluid circulation under stress [21], appears reasonable for the Thailand case.

The remagnetization direction is consistent with Thailand having been a part of Eurasia at collision time (latitude data) but having suffered some 20° of clockwise rotation subsequently (declination data), most of it probably prior to 20 Ma. This value is similar to the observed azimuthal difference of Mesozoic fold axes on either side of the Red River fault, and therefore consistent with the early geometry of and sense of motion on this fault [2,3]. Further sampling of the Jurassic and Cretaceous formations is needed to ascertain the age of the remagnetization event.

An alternative interpretation of the late Neogene data of McCabe et al. [15] suggests that

further rotation may have continued into the Neogene. Despite the unfortunate consequence of the remagnetization that paleopositions of Thailand prior to the Cenozoic cannot yet be constrained by paleomagnetic means, the event is of great interest in the frame of the India-Asia collision and validates the concept of extrusion tectonics. More accurately dated Cenozoic samples will be required before the timing of rotation can be further constrained.

Acknowledgements

This paper vindicates Neil Opdyke's early doubts as to the primary nature of the Achache and Courtillot results. Samples were collected in 1981 by J. Achache, with the help of E. Buffetaut, P. Janvier, R. Ingavat and S. Bunopas. The paper benefited from discussions with and reviews by Paul Tapponnier, Ken Hoffman, Randy Enkin and two anonymous referees.

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